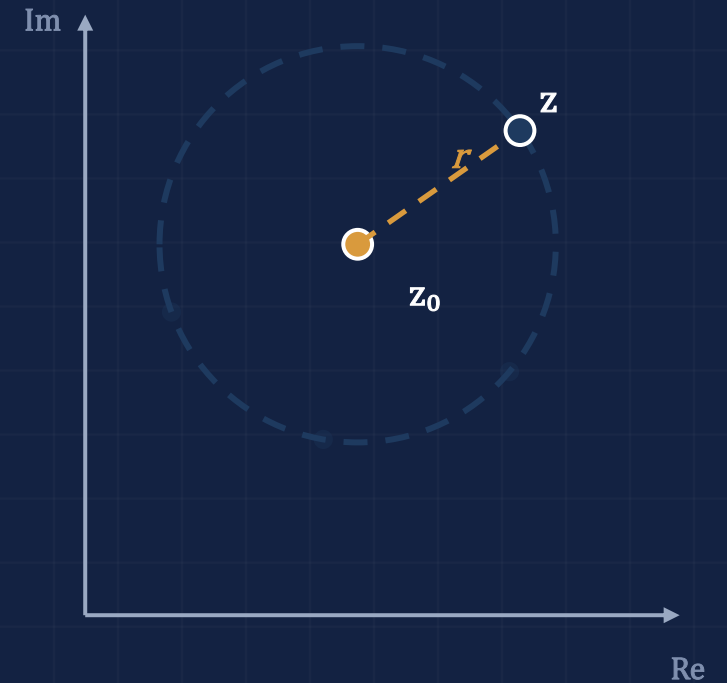


CLASS 12 • MATHEMATICS • COMPLEX NUMBERS

Geometry and Locus of Complex Numbers

Equation of a Circle in Complex Form



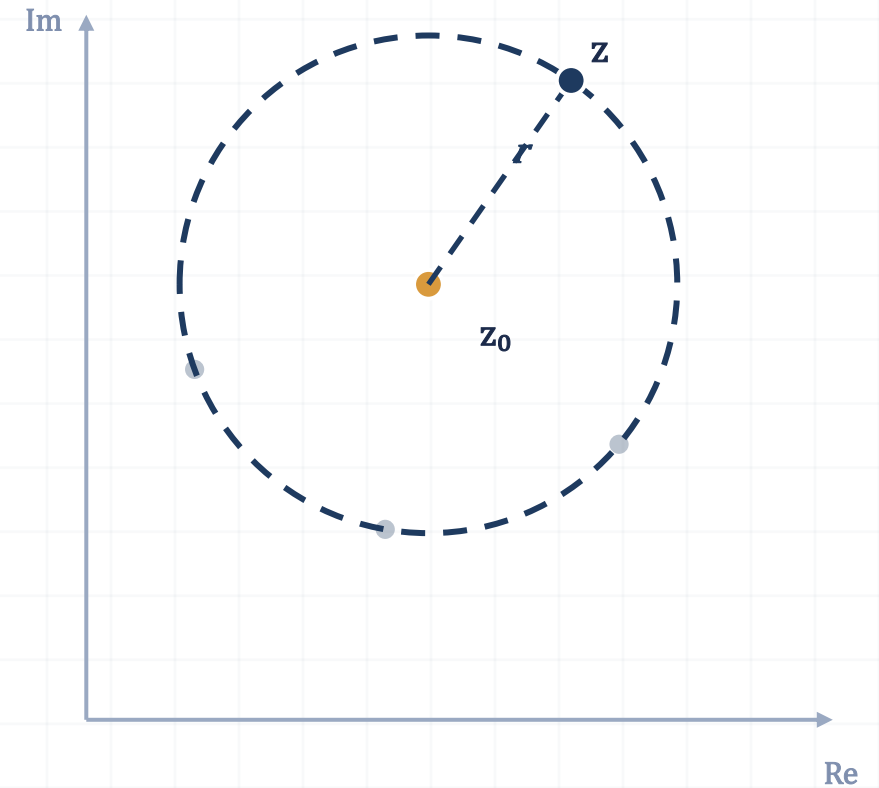
z traces a path of constant distance from z_0

Learning Objectives — What Is a Locus?

- 1 Understand the locus of a point moving in the complex plane.
- 2 Understand the equation of a circle in complex form, $|z - z_0| = r$.
- 3 Solve simple problems involving $|z - z_0| = r$.

A locus is the path traced by a point moving according to a given condition.

Constant distance from a fixed point → **Circle**



Geometrical Interpretation — The Formula

Let $z = x + iy$ and $z_0 = a + ib$

$$|z - z_0| = r$$

$z_0 \rightarrow$ Centre

$r \rightarrow$ Radius

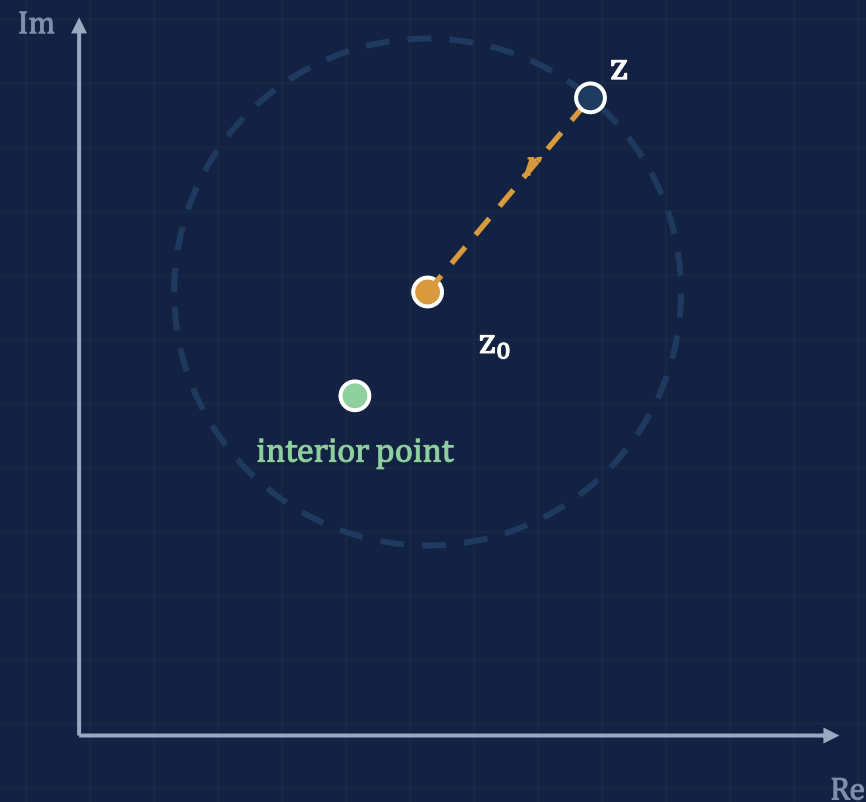
$z \rightarrow$ Any point on the circle

$$|z - z_0| < r$$

Interior of the circle

$$|z - z_0| > r$$

Exterior of the circle



Connection with Cartesian Form

$$z = x + iy$$

$$z_0 = a + ib$$

$$|z - z_0| = r$$



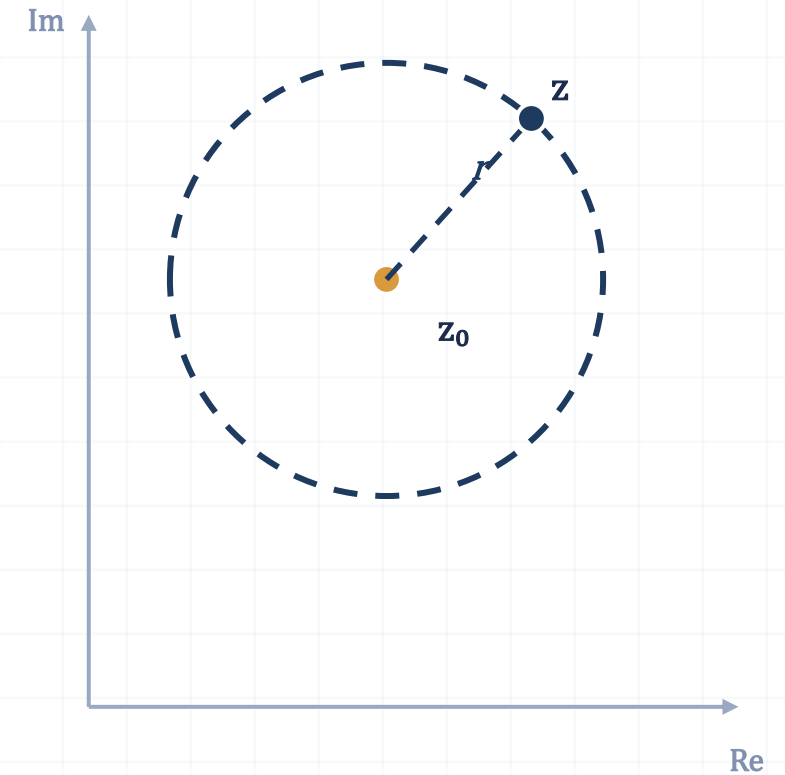
$$|(x + iy) - (a + ib)| = r$$



$$\sqrt{(x - a)^2 + (y - b)^2} = r$$

Therefore:

$$(x - a)^2 + (y - b)^2 = r^2$$



Worked Examples — Part 1

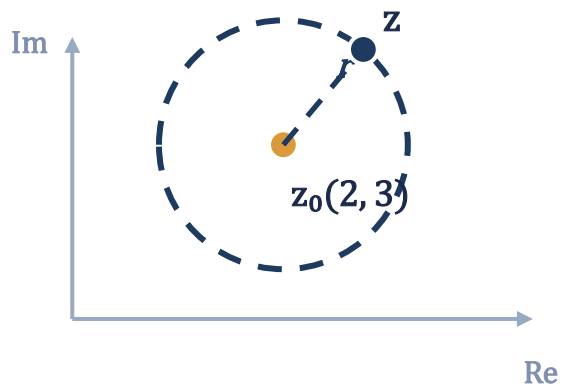
SIMPLE EXAMPLE

Find the centre and radius of $|z - (2 + 3i)| = 5$.

Compare with $|z - z_0| = r$:

$$z_0 = 2 + 3i, \quad r = 5$$

Centre = (2, 3) Radius = 5



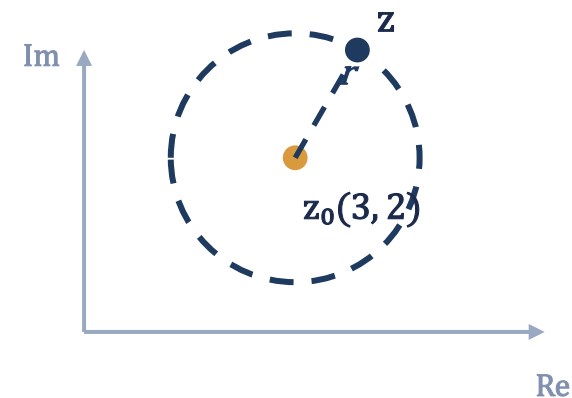
SOLVED PROBLEM 1

Find the equation of the circle with centre $3 + 2i$ and radius 4.

$$|z - z_0| = r$$

$$z_0 = 3 + 2i, \quad r = 4$$

$$|z - (3 + 2i)| = 4$$



Worked Examples — Part 2

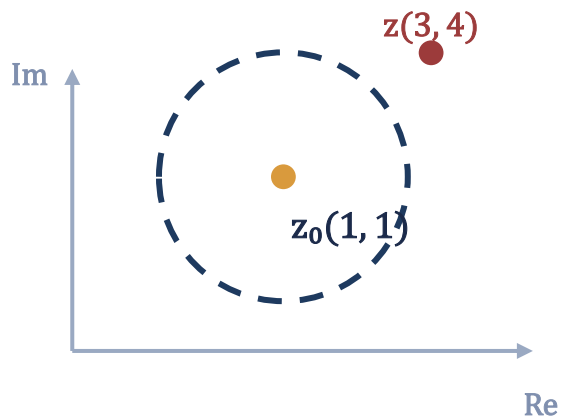
SOLVED PROBLEM 2

Does $z = 3 + 4i$ lie on the circle $|z - (1 + i)| = 4$?

$$|z - z_0| = |(3+4i) - (1+i)|$$

$$= |2 + 3i| = \sqrt{13} \approx 3.61$$

$\sqrt{13} \neq 4$ — z does NOT lie on the circle



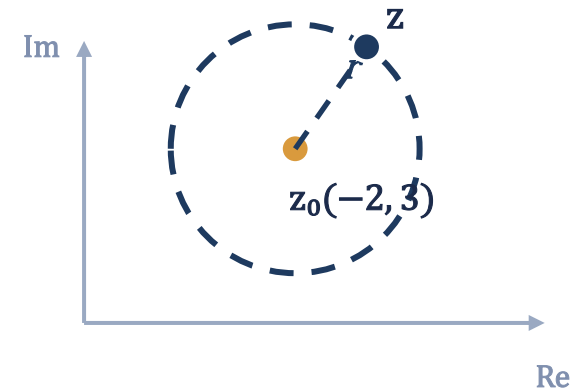
SOLVED PROBLEM 3

Find the centre and radius of $|z + 2 - 3i| = 6$.

Rewrite in standard form:

$$|z - (-2 + 3i)| = 6$$

Centre = $(-2, 3)$ Radius = 6



BEFORE WE WRAP UP

Quick Revision — Quick Check

Circle $|z - z_0| = r$

Interior $|z - z_0| < r$

Exterior $|z - z_0| > r$

Centre / Radius $z_0 = a + ib, r$

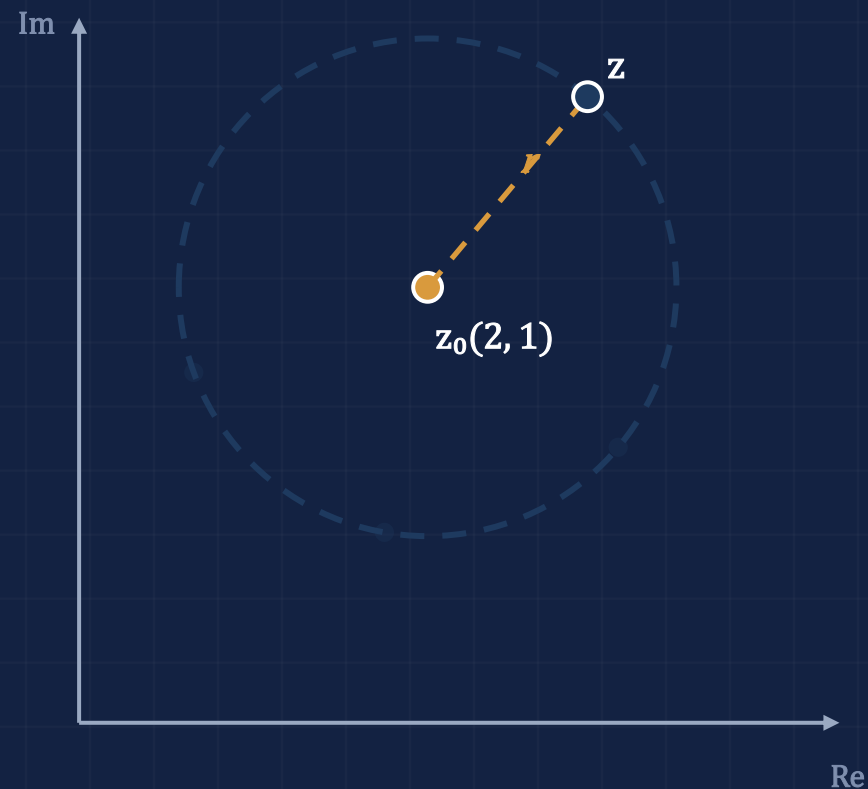
QUICK CHECK

What does $|z - (2 + i)| = 3$ represent?

A A straight line

B ✓ Circle, centre (2, 1), radius 3

C Circle, centre (3, 2), radius 1



KEY TAKEAWAY

**Fixed distance from
a fixed point → Circle**

$$|z - z_0| = r$$

Remember: z_0 is the centre and r is the radius.

